



One report, three fixes

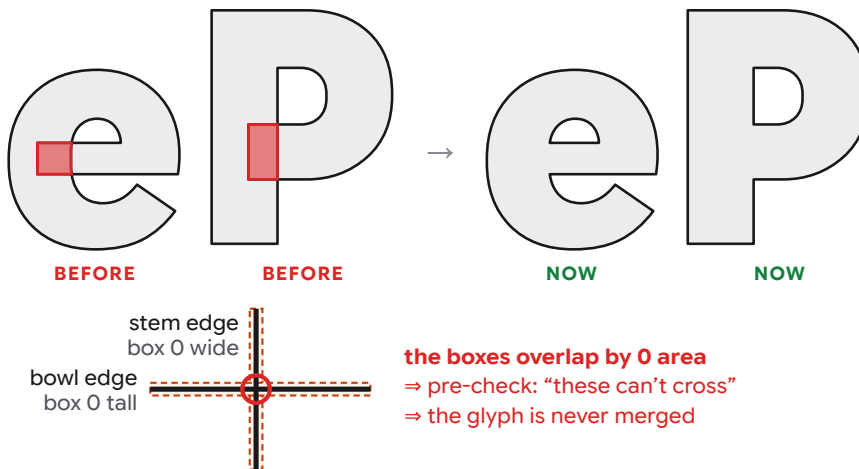
slice-web

73f9f96 → 26fd388

6 October 2026

A user outlined Google Sans Flex 4.005 Condensed Black (opz 18, wdth 80, wght 900) in Illustrator and still found overlaps. Following that up turned up two more bugs.

1 “Remove overlapping contours” left overlaps behind



Outline view, as Illustrator shows it after “Create Outlines”. Red: area filled twice, fenced by a path running through the inside of the letter.

Cause. A pre-check skipped any glyph whose segments’ bounding boxes did not overlap *by area*, and a straight stem’s box has none (left). The merge also split hairline spikes off as stray zero-area paths.

Fix. No pre-check: every glyph is merged (0.2 s for all 682). Contours that enclose nothing are dropped.

45 → 0 glyphs still overlapping (37) or carrying stray paths (8)

2 Static instances kerned like the Regular master

VARIABLE FONT AT THIS LOCATION — AND NOW

AVATAR Typeface

AVATAR Typeface

BEFORE: SAME GLYPHS, REGULAR’S KERNING — THE LINE ENDS 224 UNITS EARLY

GPOS: kern T-y
-110 (Regular’s) + pointer

GDEF variation store
Δ at this location: +86

Before: copied as-is; with no axes left, nothing computes Δ, so T-y kerns -110 at every weight.
Now: -110 + 86 = -24 written in; store removed.

20,251 → 0

of 110,224 character pairs placed differently from fontTools

394

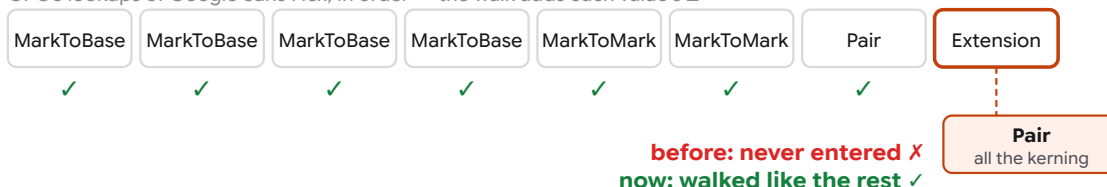
units: worst accent misplacement, where no precomposed letter exists

758

of 783 Google Fonts variable fonts carry variable kerning or anchors

3 Kerning inside “extension” lookups was never adjusted

GPOS lookups of Google Sans Flex, in order — the walk adds each value’s Δ



Cause. Compilers often wrap the biggest lookup — usually the kerning — in an extension. The walk that adds each value’s Δ treated the wrapper as empty, so partial instances that pin an axis off its default kept wrong kerning. write-fonts’ **RemapVarStore** had the same blind spot until 0.54.0.
Fix. The walk unwraps extensions.

14,298 → 0

pairs wrong at every weight sampled (wdth 80, wght 400:900)

250

of 783 Google Fonts variable fonts keep their kerning in an extension

Checked by engines that are not ours.

Overlaps with skia-pathops (what fontTools uses). Positioning by shaping every character pair with HarfBuzz, against fontTools’ instance and the variable font itself.

Beyond this font.

Ten more Google Fonts, sampled by survey: 8 had static kerning wrong by 35,173–157,980 pairs; none now. Conformance corpus 302/302, and the original Slice passes the new case too.

Reproduce.

`tools/overlap-check.py`, `kerning-compare.py`, `kerning-sample.sh`, `gdef-store-survey.py`; results in `tools/README.md`. Set in Google Sans Flex, sliced by the fixed Slice.